

13 — Insight Rule Registry

Status: DRAFT v0.1 (2026-06-12)

The rules that find risks, opportunities, anomalies, and drivers in the data.

Each rule is a deterministic check. Cards only name the rule ids they show.

Purpose

An insight is a fact worth attention, like "target achievement is below 80%".

A rule defines:

- When does this insight fire? (the trigger)
- How bad / how big is it? (the impact)
- Where do we look for the likely reason? (the evidence)
- What can the user do about it? (the actions)

Rules are deterministic. They run on Cube.js numbers. The LLM only writes the words around the result. The LLM never decides if a rule fired.

Design rules

- One rule = one entry = one id.
- The rule lives here, once. Cards just list which rule ids they show.
- Thresholds live in the rule as a default. A tenant may override them (see 20).
- Confidence and priority are computed by the engine, not by the LLM (see below).
- actions here are ADVISORY defaults. The persona card (10) owns the role-appropriate insight→action mapping via actions[].from_insight — that is what a user actually sees.

Insight types

type	meaning
risk	something going wrong (target miss, decline, stockout)
opportunity	something to gain (fast SKU, under-distributed product)
anomaly	an odd jump or drop versus the normal pattern
driver	what is moving a number (volume, price, mix, distribution)

Trigger kinds

A rule fires using one of three trigger kinds.

- threshold

→ metric compared to a fixed number

(e.g. target_ach < 80)
- variance

→ metric compared to a baseline period

(e.g. sales down >10% vs last month)
- anomaly

→ metric far from its history average

(e.g. z-score > 2 vs L3M avg)

Field table

Field	Type	Required	Notes
rule_id	string (snake_case)	yes	stable id, e.g. target_gap
type	enum risk opportunity anomaly driver	yes	the insight type
title	string	yes	short plain title shown to user
trigger.kind	enum threshold variance anomaly	yes	how the rule fires
trigger.metric	string (metric id)	yes	the metric being checked; must exist in 12
trigger.operator	enum lt lte gt gte eq ne	threshold only	comparison vs value
trigger.value	number	threshold only	the fixed number
trigger.baseline	enum previous_month previous_quarter smly target	variance only	what to compare against
trigger.direction	enum up down	variance only	which way of change matters
trigger.min_change_pct	number	variance only	minimum % change to fire
trigger.history_window	enum 13m 16m 113m	anomaly only	the history average to use
trigger.z_threshold	number	anomaly only	how far from normal to fire
breakdown	string[]	yes	dims to slice to find where it is worst

Field	Type	Required	Notes
impact.metric	string (metric id)	yes	metric that sizes the impact (for priority)
evidence	string[]	no	other rule ids to check for the likely reason
actions	string[]	no	ADVISORY default action ids. The authoritative, role-appropriate action a user sees comes from the persona card (10) via actions[].from_insight .
applies_to.operating_grain	string[]	no	which role grains this rule is relevant for

JSON Schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://cpg-analytics/spec/insight-rule.schema.json",
  "title": "InsightRuleRegistry",
  "type": "array",
  "items": {
    "type": "object",
    "additionalProperties": false,
    "required": ["rule_id", "type", "title", "trigger", "breakdown", "impact"],
    "properties": {
      "rule_id": { "type": "string", "pattern": "^[a-z][a-z0-9_]*$" },
      "type": { "enum": ["risk", "opportunity", "anomaly", "driver"] },
      "title": { "type": "string", "minLength": 1 },

      "trigger": {
        "type": "object",
        "additionalProperties": false,
        "required": ["kind", "metric"],
        "properties": {
          "kind": { "enum": ["threshold", "variance", "anomaly"] },
          "metric": { "type": "string" },
          "operator": { "enum": ["lt", "lte", "gt", "gte", "eq", "ne"] },
          "value": { "type": "number" },
          "baseline": { "enum": ["previous_month", "previous_quarter", "smly", "target"] },
          "direction": { "enum": ["up", "down"] },
          "min_change_pct": { "type": "number" },
          "history_window": { "enum": ["13m", "16m", "113m"] },
          "z_threshold": { "type": "number" }
        }
      },
      "allOf": [
        { "if": { "properties": { "kind": { "const": "threshold" } } },
          "then": { "required": ["operator", "value"] } },
        { "if": { "properties": { "kind": { "const": "variance" } } },
          "then": { "required": ["baseline", "direction", "min_change_pct"] } },
        { "if": { "properties": { "kind": { "const": "anomaly" } } },
          "then": { "required": ["history_window", "z_threshold"] } }
      ]
    },

    "breakdown": { "type": "array", "items": { "type": "string" } },
    "impact": {
      "type": "object",
      "additionalProperties": false,
      "required": ["metric"],
      "properties": { "metric": { "type": "string" } }
    },

    "evidence": { "type": "array", "items": { "type": "string" } },
    "actions": { "type": "array", "items": { "type": "string" } },
    "applies_to": {
```

```

    "type": "object",
    "additionalProperties": false,
    "properties": { "operating_grain": { "type": "array", "items": { "type": "string" } } }
  }
}
}
}

```

Confidence and priority (engine computes both — never the LLM)

The rule says *what* to check. The engine then scores the result:

Confidence (how reliable is this insight?)

- └─ data completeness (are all needed metrics + periods present?)
- └─ evidence strength (does the signal show across breakdown dims?)
- └─ stability (is it more than one-day noise vs the baseline?)

Priority (how important is it to act?)

- └─ impact value (from impact.metric)
- └─ urgency (month-end near? stockout active now?)
- └─ scope (national vs one outlet)
- └─ role relevance (is it in this role's job + grain?)

The engine maps the priority score into buckets **critical** | **high** | **medium** | **low**
(these are the values the home layout filters on via `filters.priority` in 14).

Weights for these are engine config (a tenant may tune them — see 20).

The LLM may explain a score in words. It must never set the number.

Validation rules (check in CI)

1. Every `rule_id` is unique.
2. `trigger.metric` and `impact.metric` exist in the metric registry (12).
3. Every id in `evidence` is an existing `rule_id`.
4. `actions` are advisory. Every id SHOULD be defined on at least one persona card (10);
what a user sees is resolved from THEIR card's `actions[].from_insight`. (No separate action map.)
5. `breakdown` and `applies_to.operating_grain` dims are valid Cube.js dimensions.
6. Trigger fields match the `kind` (enforced by the JSON Schema `allOf`).

Worked example

See `examples/insight-rules.yaml` . One rule:

```
- rule_id: target_gap
  type: risk
  title: "Target achievement is below expected level"
  trigger:
    kind: threshold
    metric: target_achievement_pct
    operator: lt
    value: 80
  breakdown: [region, distributor, sku]
  impact:
    metric: sales_gap_value
  evidence: [distributor_underperformance, outlet_coverage_drop, stockout_risk]
  actions: [analyze_target_gap, review_weak_distributors] # advisory; card decides per role
  applies_to:
    operating_grain: [nation, region, area]
```